

Machine Learning

Lecture 3: Linear Regression

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本节内容

- Linear regression
- Normal equation
- Maximum Likelihood Estimation
- MSE v.s. MAE

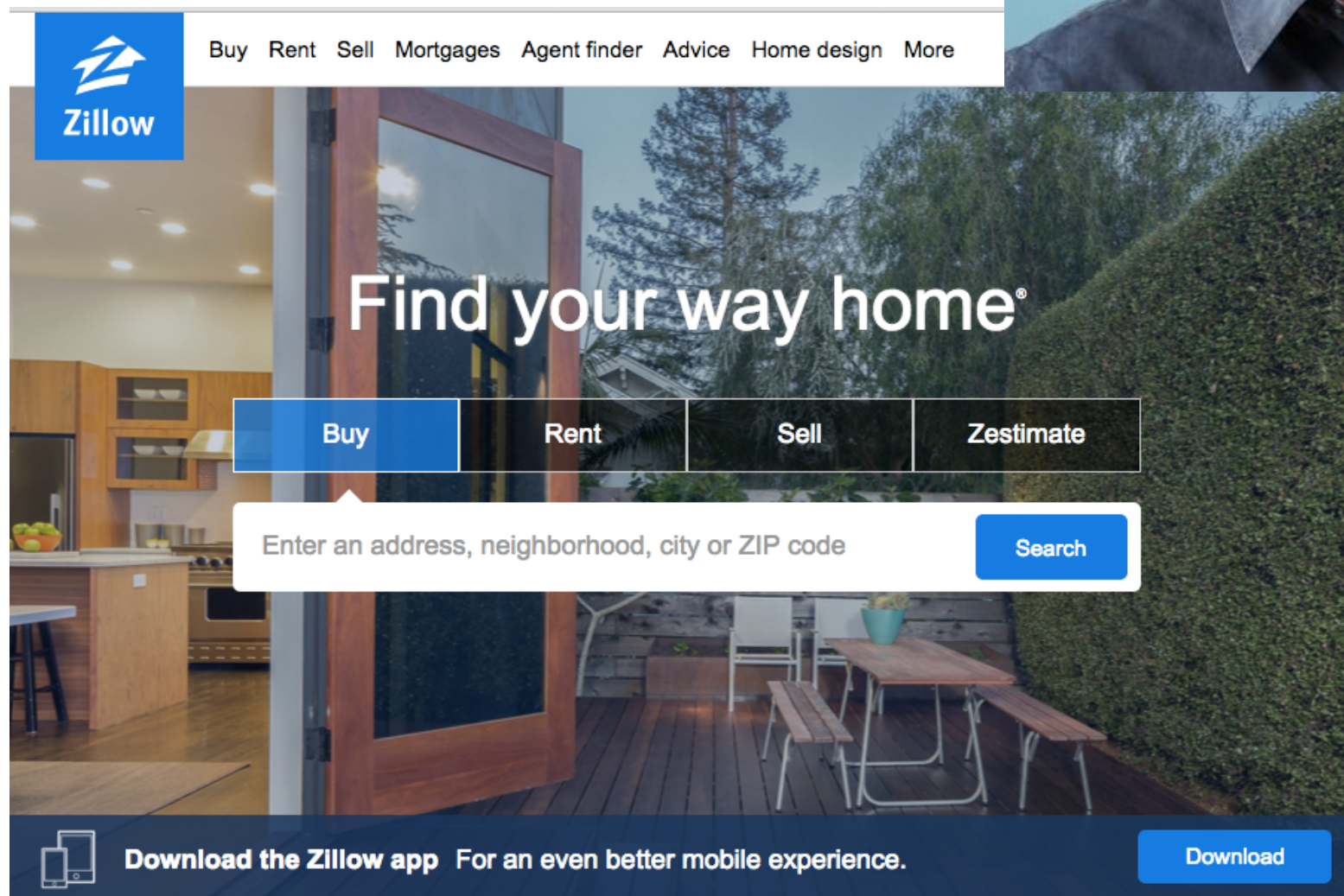
1. 线性回归 (linear regression)

Example

Rich Barton

Born June 2, 1967.

Founder of Expedia, Zillow,
Glassdoor and Trover



Net worth US \$400 million

In 2002, named as one of top innovators under 35 by *MIT Technology Review*

In April 2014, named to Barack Obama's *Presidential Ambassador for Global Entrepreneurship*

Housing prices

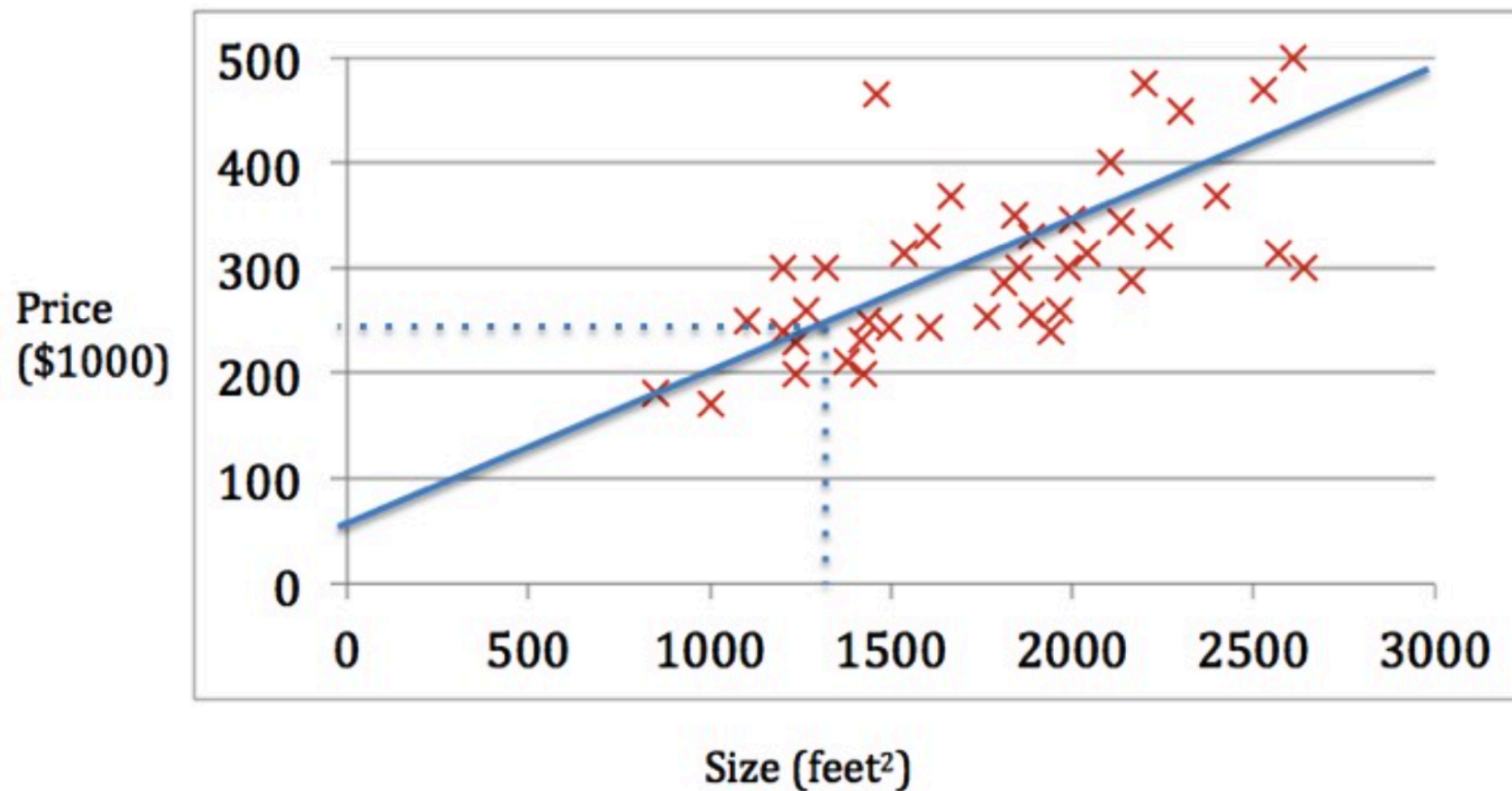
Training set of housing prices (Portland, OR)

Size in feet ²	Price in \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

Notation:

- m : number of training examples
- x : “Input” variables/features
- y : “Output” variables
- (x_i, y_i) : i -th training example (i -th datum)
- $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^m$: the training set

Linear regression problems with one variable



supervised learning

Given the “right output” (labels)
for each example

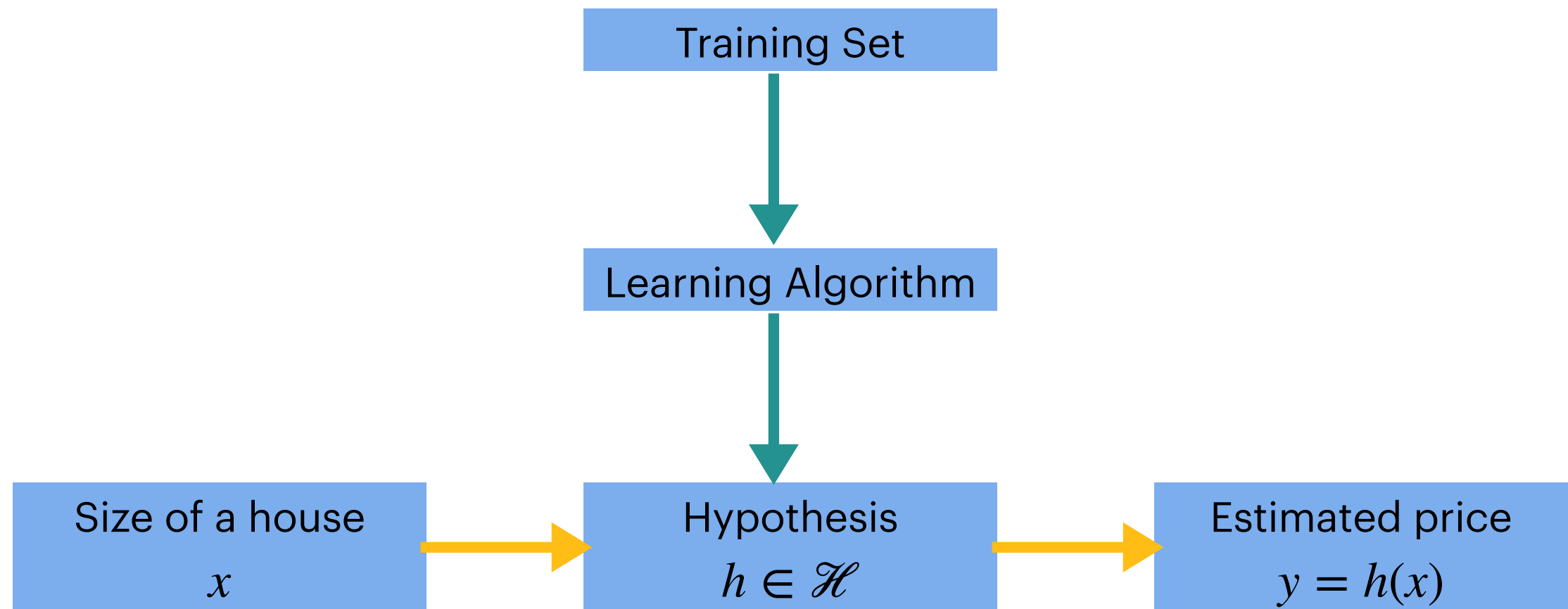
Regression problems

Predict real-valued (continuous) output

Classification problems

Predict categorical (discrete) output

Learning process



- What is our hypothesis class $\mathcal{H}$?

Linear regression problems with one variable

How do we represent h ?

- Linear (affine) function

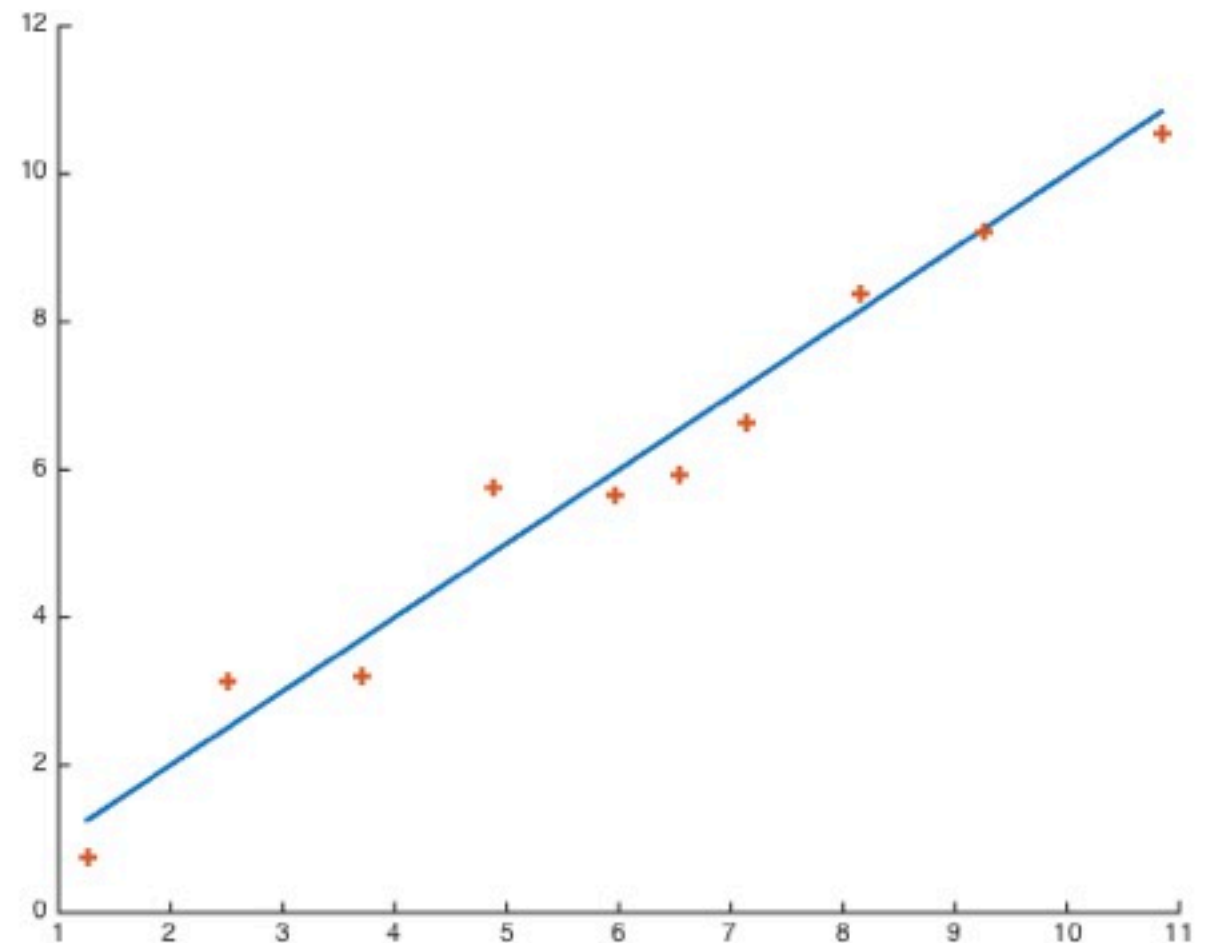
$$h(\cdot; \theta) : \mathcal{X} \rightarrow \mathcal{Y}$$

$$h(x; \theta) = \theta_0 + \theta_1 x$$

$$\theta = [\theta_0, \theta_1]^T$$

- Hypothesis class

$$\mathcal{H} = \{ \theta_0 + \theta_1 x \mid \theta_0, \theta_1 \in \mathbb{R} \}$$



How do we choose the optimal h ?

Construct loss function

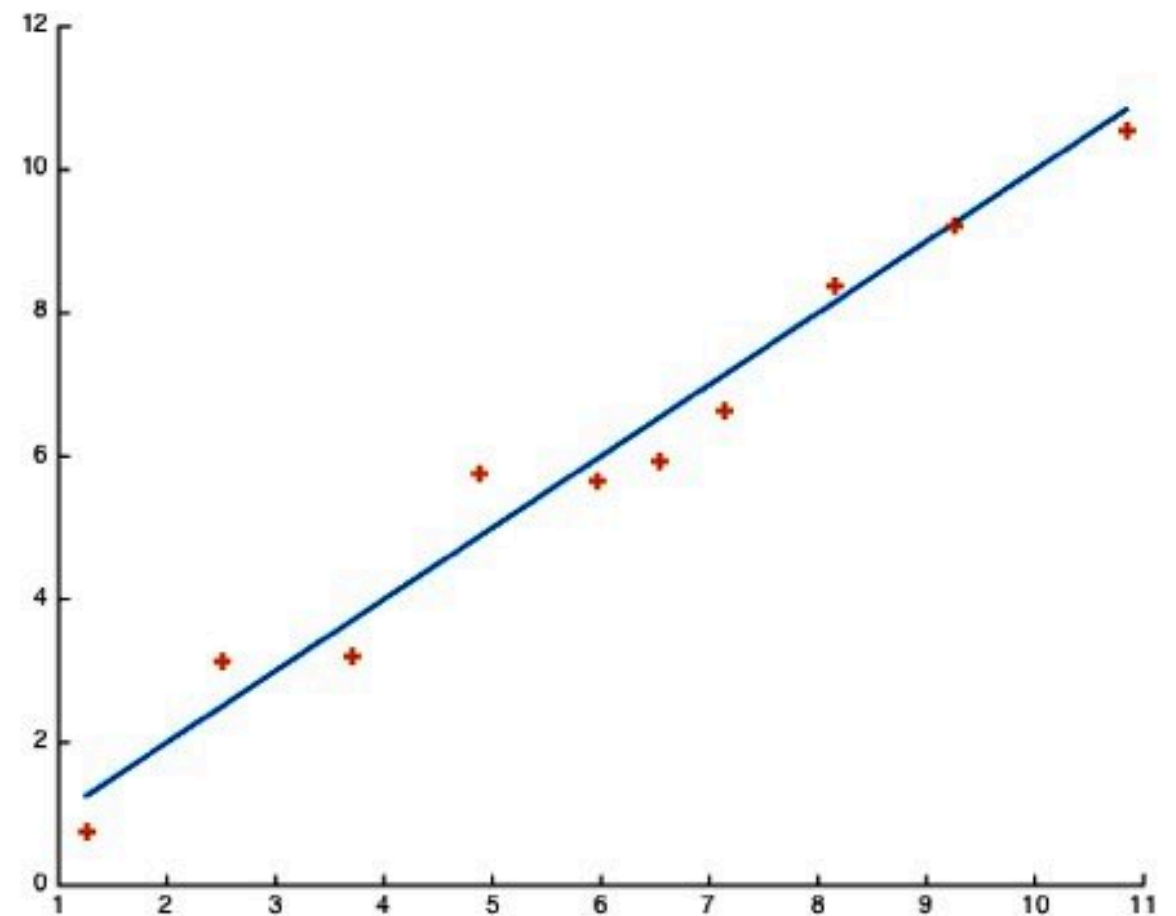
How much loss/cost is generated by this h on training set of housing prices (Portland, OR)?

How do we represent h ?

- The prediction of $h(x; \theta)$ for the i -th datum

$$h(x_i; \theta) = \theta_0 + \theta_1 x_i$$

- The “real” output y_i
- The “error” between the predicted and the “real” output
 $(h(x_i; \theta) - y_i)^2 = (\theta_0 + \theta_1 x_i - y_i)^2$



📖 Do we have to choose squared error?

Construct loss function

- The (averaged) total error on the training set

$$J(h(\theta_0, \theta_1), \mathcal{D}) = \frac{1}{2m} \sum_{i=1}^m (h(x_i; \theta_0, \theta_1) - y_i)^2$$

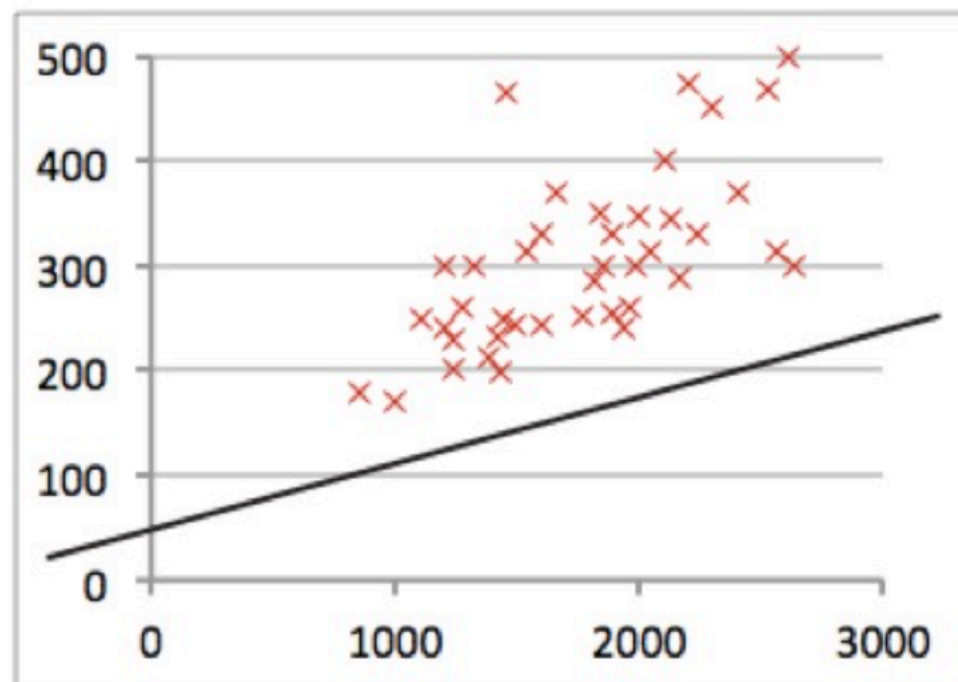
- Intuitively, h should minimize the total error $J(h(\theta_0, \theta_1), \mathcal{D})$ over the training set $\mathcal{D}$
- Choose (θ_0, θ_1) as the solution of

$$\min_{\theta_0, \theta_1} J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h(x_i; \theta_0, \theta_1) - y_i)^2$$

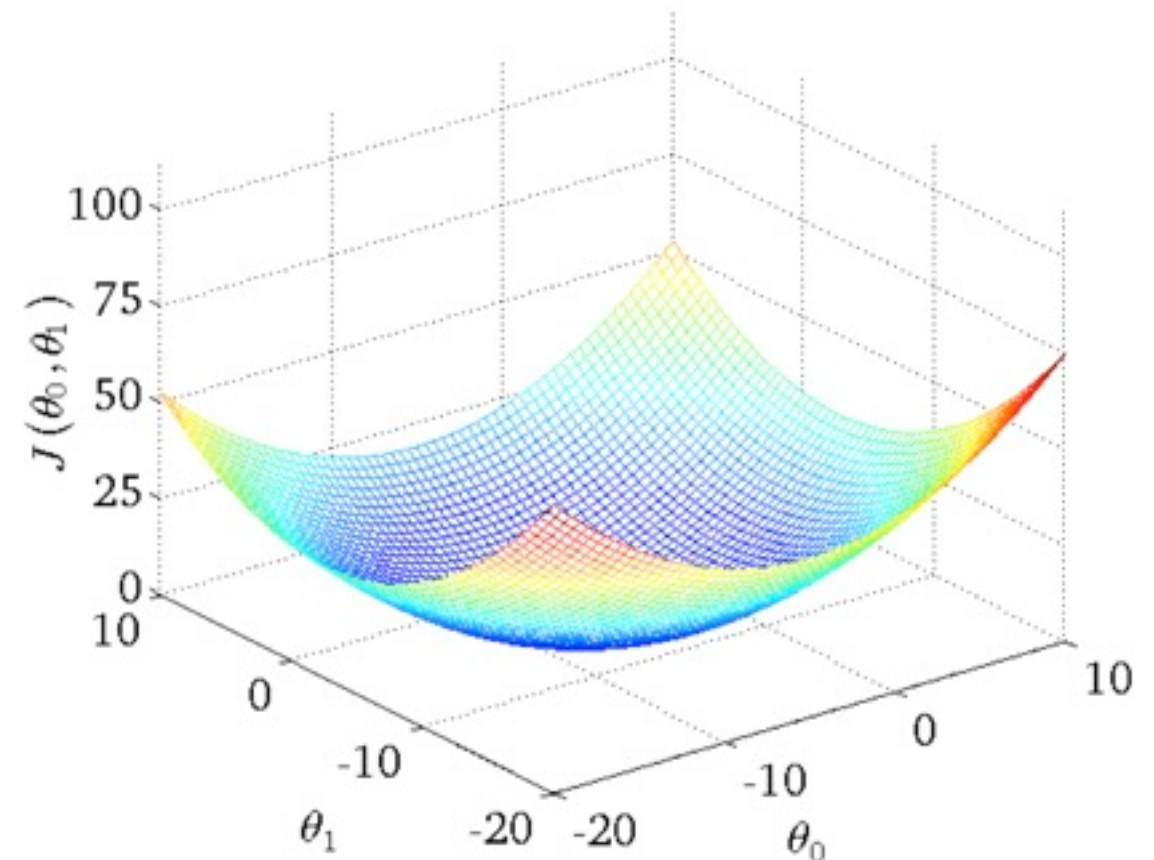
Finding optimal parameter

$$\min_{\theta_0, \theta_1} J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h(x_i; \theta_0, \theta_1) - y_i)^2$$

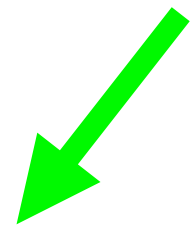
- For fixed parameters, $h(x, \theta_0, \theta_1)$ is a linear function of x
- For given training set $\mathcal{D}$, $J(\theta_0, \theta_1)$ is quadratic of (θ_0, θ_1)



$$h(x; 50, 0.06) = 50 + 0.06x$$



Linear regression with multiple variable



Size (feet ²)	Number of bedrooms	Number of floors	Age of home (years)	Price (\$1000)	Zipcode
2104	5	1	45	460	18015
1416	3	2	40	232	18014
1534	3	2	30	315	18323
852	2	1	36	178	18001
...	...	...	...		

- Multiple variables

$$h(\mathbf{x}) = h(\mathbf{x}; \boldsymbol{\theta}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

- For convenience of notion, define

$$x_0 = 1, \quad \mathbf{x} = [x_0, x_1, \dots, x_n]^T$$

- Example:

$$h(\mathbf{x}) = 80x_0 + 0.1x_1 + 0.01x_2 + \dots + 3x_3 - 2x_4$$

Dummy coding

- ▶ Continuous variables (连续变量) : Real-valued numbers
Examples: 23; 1000.999
- ▶ Ordinal variables (有序类别变量) : A variable that can take on one of a number of possible values, thus assigning each individual to a particular group or “category”, and is ordered or ranked
Examples: integer 1, 2, 3, 4...; “strongly disagree, disagree, neutral, agree, strongly agree”
- ▶ Categorical variables ((无序) 类别变量)
A variable that can take on one of a limited, and usually fixed, number of possible values, thus assigning each individual to a particular group or “category”, but is NOT ordered or ranked
Examples: “yes, no”; “female, male”; zipcode

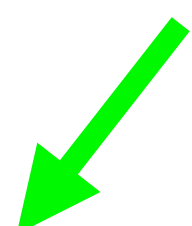
Dummy coding

District	Price 1000's (y)
徐汇	100
浦东	90
闵行	80
浦东	30
浦东	70
⋮	⋮

One way of coding:

徐汇	闵行	浦东	Price 1000's (y)
1	0	0	100
0	0	1	90
0	1	0	80
0	0	1	30
0	0	1	70
⋮	⋮		

Nonlinear Transformation: “Linear” w.r.t. parameters



Size (feet ²)	Number of bedrooms	Number of floors	Age of home (years)	Price (\$1000)	Zipcode
2104	5	1	45	460	18015
1416	3	2	40	232	18014
1534	3	2	30	315	18323
852	2	1	36	178	18001
...	...	...	...		

Linear models:

$$h(\mathbf{x}) = h(\mathbf{x}; \boldsymbol{\theta}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

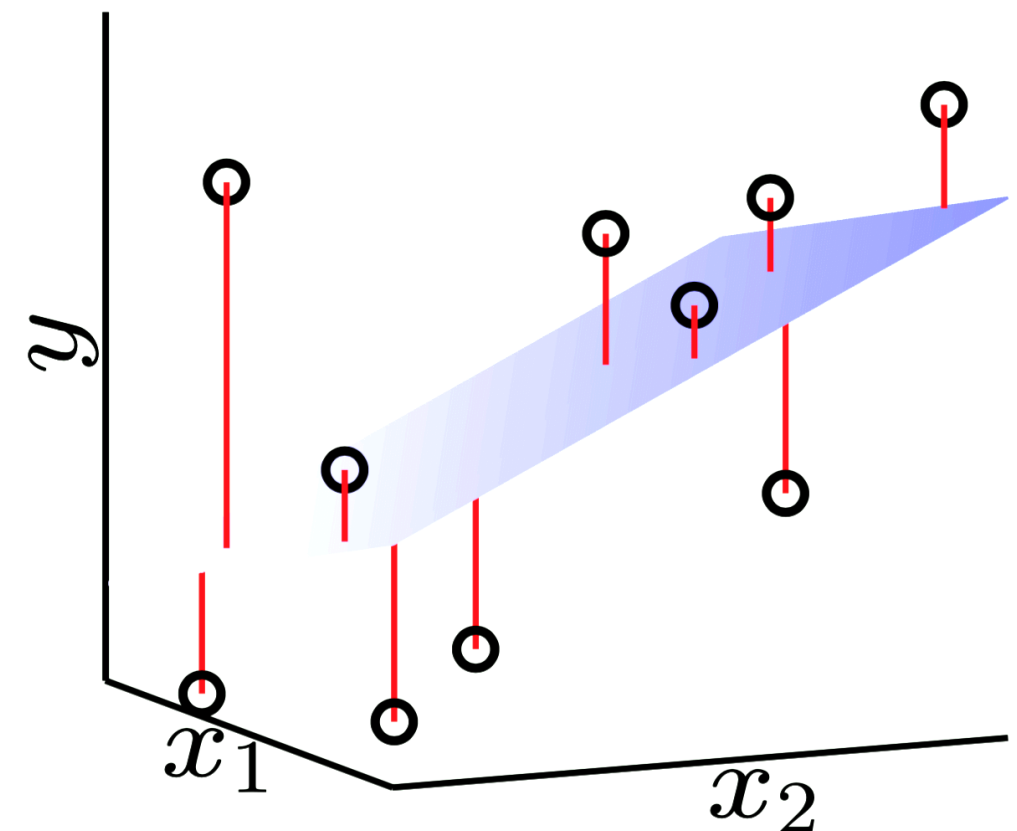
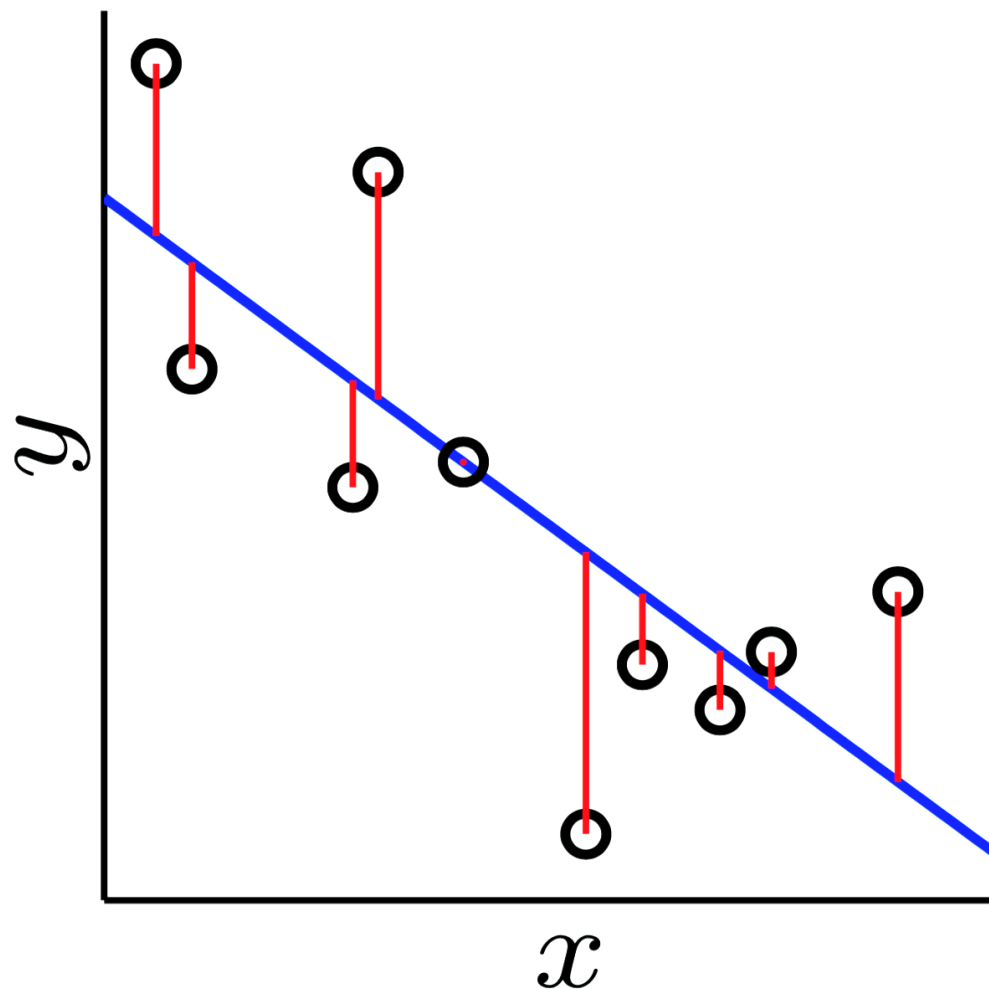
$$h(\mathbf{x}) = h(\mathbf{x}; \boldsymbol{\theta}) = \theta_0 + \theta_1 x_1 x_2 + \theta_2 x_1^2 (x_2 - x_3)$$

Linear regression with multiple variable

- Multivariate linear regression

$$h(\mathbf{x}) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

Illustration of linear regression



Linear regression with multiple variable

► Notation:

$$\mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_m^T \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y^m \end{bmatrix}$$

$$\mathbf{x}_i = [x_{i,0}, x_{i,1}, x_{i,2}, \dots, x_{i,n}]^T$$

$$x_{i,j}, i = 1, \dots, m; j = 1, \dots, n$$

- Hypothesis: $h(\mathbf{x}) = \boldsymbol{\theta}^T \mathbf{x}$
- Hypothesis class: $\mathcal{H} = \{h(\mathbf{x}) \mid \boldsymbol{\theta} \in \mathbb{R}^{n+1}\}$
- Parameters/Cost function variables: $\boldsymbol{\theta}$
- Cost function: $J(\boldsymbol{\theta}) = \frac{1}{2} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$ with gradient

$$\nabla J(\boldsymbol{\theta}) = \mathbf{X}^T (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})$$

Finding the optimal parameters: Gradient Descent Method

- ▶ Batch Gradient: $\nabla J(\boldsymbol{\theta}) = \sum_{i=1}^m (\mathbf{x}_i^T \boldsymbol{\theta} - y_i) \mathbf{x}_i$
- ▶ Batch Gradient Descent: until convergence, repeat

$$\boldsymbol{\theta}^{k+1} \leftarrow \boldsymbol{\theta}^k - \alpha^k \nabla J(\boldsymbol{\theta}^k)$$

2. Normal Equation

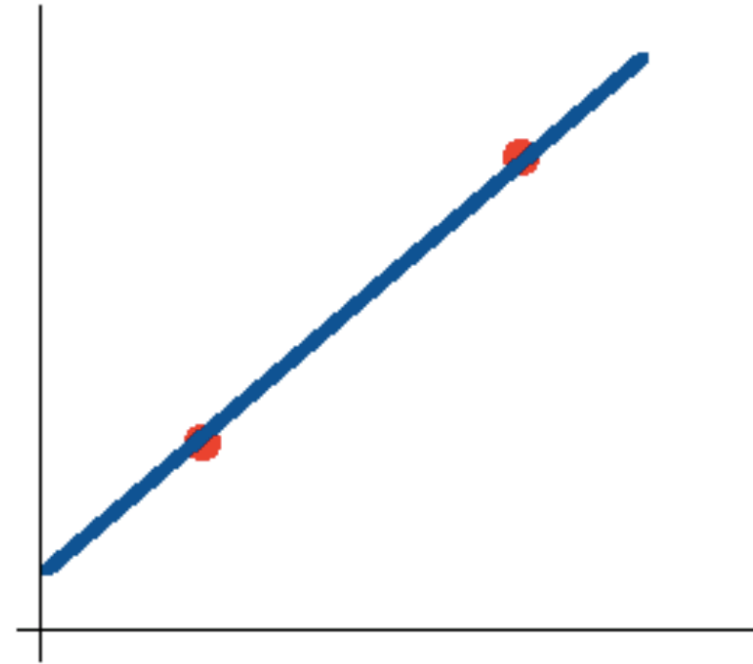
MSE (均方差) and SSE (和方差)

- ▶ Consider a simple problem

- One feature, two data points
- Two unknowns: θ_0, θ_1
- Two equations

$$y_1 = \theta_0 + \theta_1 x_1 \quad (4.1)$$

$$y_2 = \theta_0 + \theta_1 x_2 \quad (4.2)$$



- ▶ Can solve this system directly

$$\mathbf{X}\boldsymbol{\theta} = \mathbf{y} \quad \rightarrow \quad \hat{\boldsymbol{\theta}} = \mathbf{X}^{-1}\mathbf{y}$$

- ▶ However, most of the time, $m > n$

- ▶ There may be no linear function that hits all the data exactly
- ▶ Instead, solve directly for minimum of MSE function

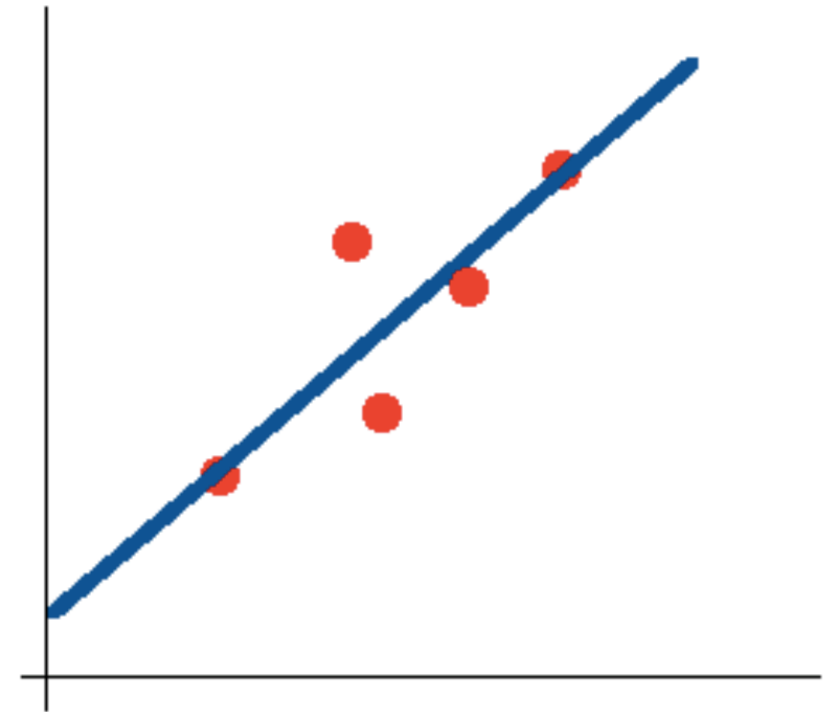
Normal equation (法方程)

- ▶ Reordering, we have

$$\mathbf{X}^T \mathbf{X} \boldsymbol{\theta} - \mathbf{X}^T \mathbf{y} = 0$$

$$\mathbf{X}^T \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^T \mathbf{y}$$

$$\boldsymbol{\theta} = (\mathbf{X}^T \mathbf{X})^+ \mathbf{X}^T \mathbf{y}$$



- ▶ $(\mathbf{X}^T \mathbf{X})^+$ is called the “pseudo-inverse” of $\mathbf{X}^T \mathbf{X}$ (伪逆矩阵)
- ▶ If $\mathbf{X}^T \mathbf{X}$ has full rank, this is the inverse $(\mathbf{X}^T \mathbf{X})^+ = (\mathbf{X}^T \mathbf{X})^{-1}$
- ▶ If $m > n$, overdetermined; gives minimum MSE fit

A^+ is called the pseudo-inverse of A if it satisfies:

- ▶ $AA^+A = A$
- ▶ $(AA^+)^T = AA^+$
- ▶ $A^+AA^+ = A^+$
- ▶ $(A^+A)^T = A^+A$

Examples: $m = 4$.

	Size (feet ²)	Number of bedrooms	Number of floors	Age of home (years)	Price (\$1000)
x_0	x_1	x_2	x_3	x_4	y
1	2104	5	1	45	460
1	1416	3	2	40	232
1	1534	3	2	30	315
1	852	2	1	36	178

$$X = \begin{bmatrix} 1 & 2104 & 5 & 1 & 45 \\ 1 & 1416 & 3 & 2 & 40 \\ 1 & 1534 & 3 & 2 & 30 \\ 1 & 852 & 2 & 1 & 36 \end{bmatrix}$$

$$y = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}$$

$$\theta = (X^T X)^{-1} X^T y$$

Examples: $m = 5$.

	Size (feet ²)	Number of bedrooms	Number of floors	Age of home (years)	Price (\$1000)
x_0	x_1	x_2	x_3	x_4	y
1	2104	5	1	45	460
1	1416	3	2	40	232
1	1534	3	2	30	315
1	852	2	1	36	178
1	3000	4	1	38	540

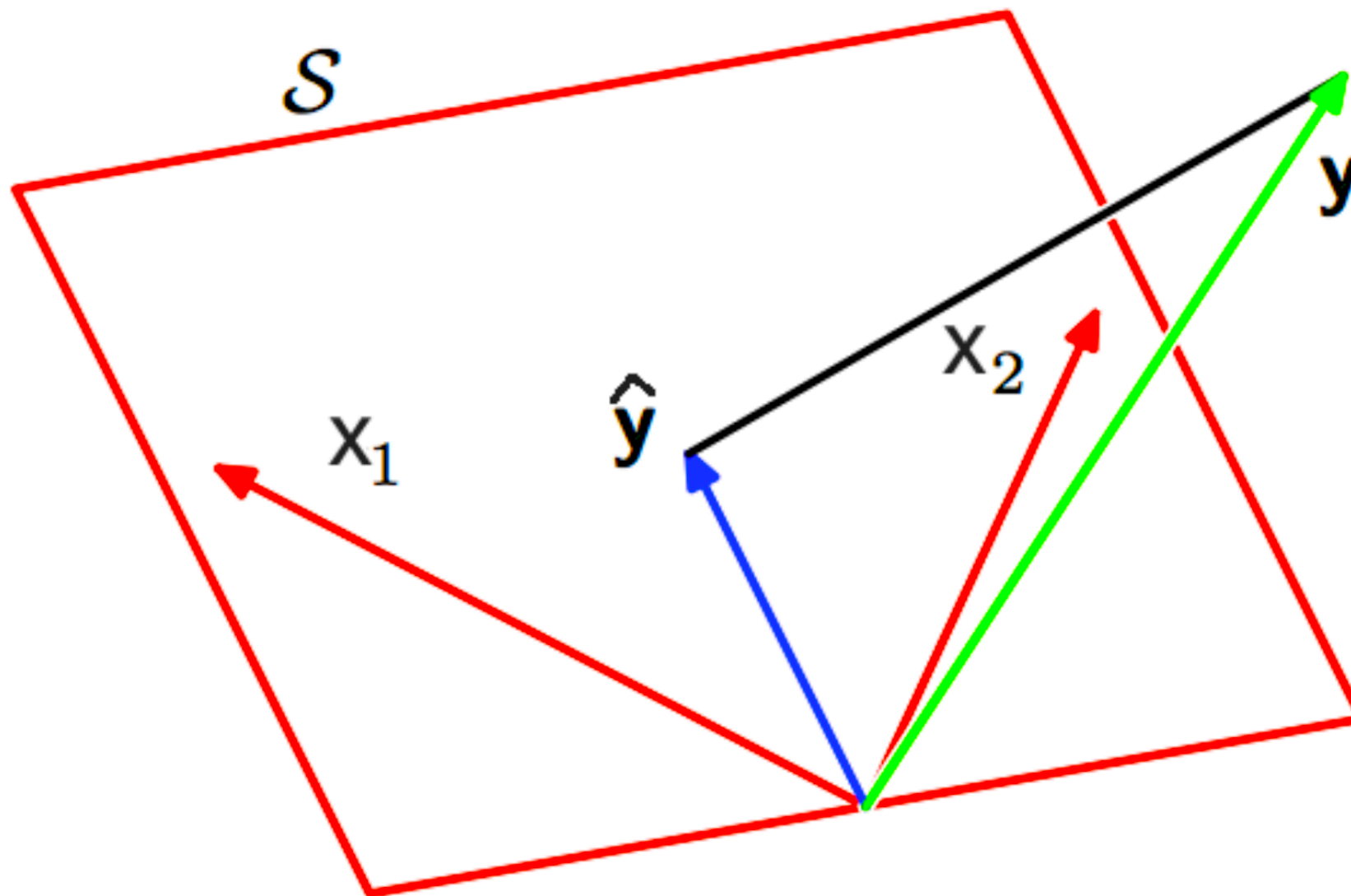
$$X = \begin{bmatrix} 1 & 2104 & 5 & 1 & 45 \\ 1 & 1416 & 3 & 2 & 40 \\ 1 & 1534 & 3 & 2 & 30 \\ 1 & 852 & 2 & 1 & 36 \\ 1 & 3000 & 4 & 1 & 38 \end{bmatrix}$$

$$y = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \\ 540 \end{bmatrix}$$

$$\Theta = (X^T X)^{-1} X^T y$$

Interpretation

- ▶ $\mathbf{X}\boldsymbol{\theta} - \mathbf{y}$ is the vector of errors in each example
- ▶ $\mathbf{X}$ are the features we have to work with for each example
- ▶ Inner product = 0: orthogonal (垂直)



3. MLE

The principle of maximum likelihood

Example

Suppose that $X^1, X^2, \dots, X^m$ are i.i.d. discrete random variables, such that $X^i \sim \text{Poisson}(\theta)$ with a pmf (probability mass function) defined as

$$\Pr(X^i = x) = \frac{\exp(-\theta)\theta^x}{x!}$$

where θ is an unknown parameter to estimate.

The principle of maximum likelihood

Question: What is the probability of observing the particular sample $\{x^1, \dots, x^m\}$? assuming that a Poisson distribution with yet unknown parameter θ generates the data?

This probability is equal to

$$Pr((X^1 = x^1) \cap \dots \cap (X^m = x^m))$$

The principle of maximum likelihood

Since the variables X^i are i.i.d., this joint probability is equal to the product of the marginal probabilities

$$Pr((X^1 = x^1) \cap \dots \cap (X^m = x^m)) = \prod_{i=1}^m Pr(X^i = x^i)$$

Given the pmf of the Poisson distribution, we have:

$$\begin{aligned} Pr((X^1 = x^1) \cap \dots \cap (X^m = x^m)) &= \prod_{i=1}^m \frac{\exp(-\theta)\theta^{x^i}}{x^i!} \\ &= \exp(-\theta m) \frac{\theta^{\sum_{i=1}^m x^i}}{\prod_{i=1}^m x^i!} \end{aligned}$$

The principle of maximum likelihood

This joint probability is a function of θ (the unknown parameter) and corresponds to the likelihood of the sample $\{x^1, \dots, x^m\}$ denoted by

$$L_m(\theta; x^1, \dots, x^m) = Pr((X^1 = x^1) \cap \dots \cap (X^m = x^m))$$

with

$$L_m(\theta; x^1, \dots, x^m) = \exp(-\theta m) \frac{\theta^{\sum_{i=1}^m x^i}}{\prod_{i=1}^m x^i!}$$

in our example.

The principle of maximum likelihood

Example

Let us assume that for $m = 10$, we have a realization of the sample equal to $\{5, 0, 1, , 1, 0, 3, 2, 3, 4, 1\}$, then

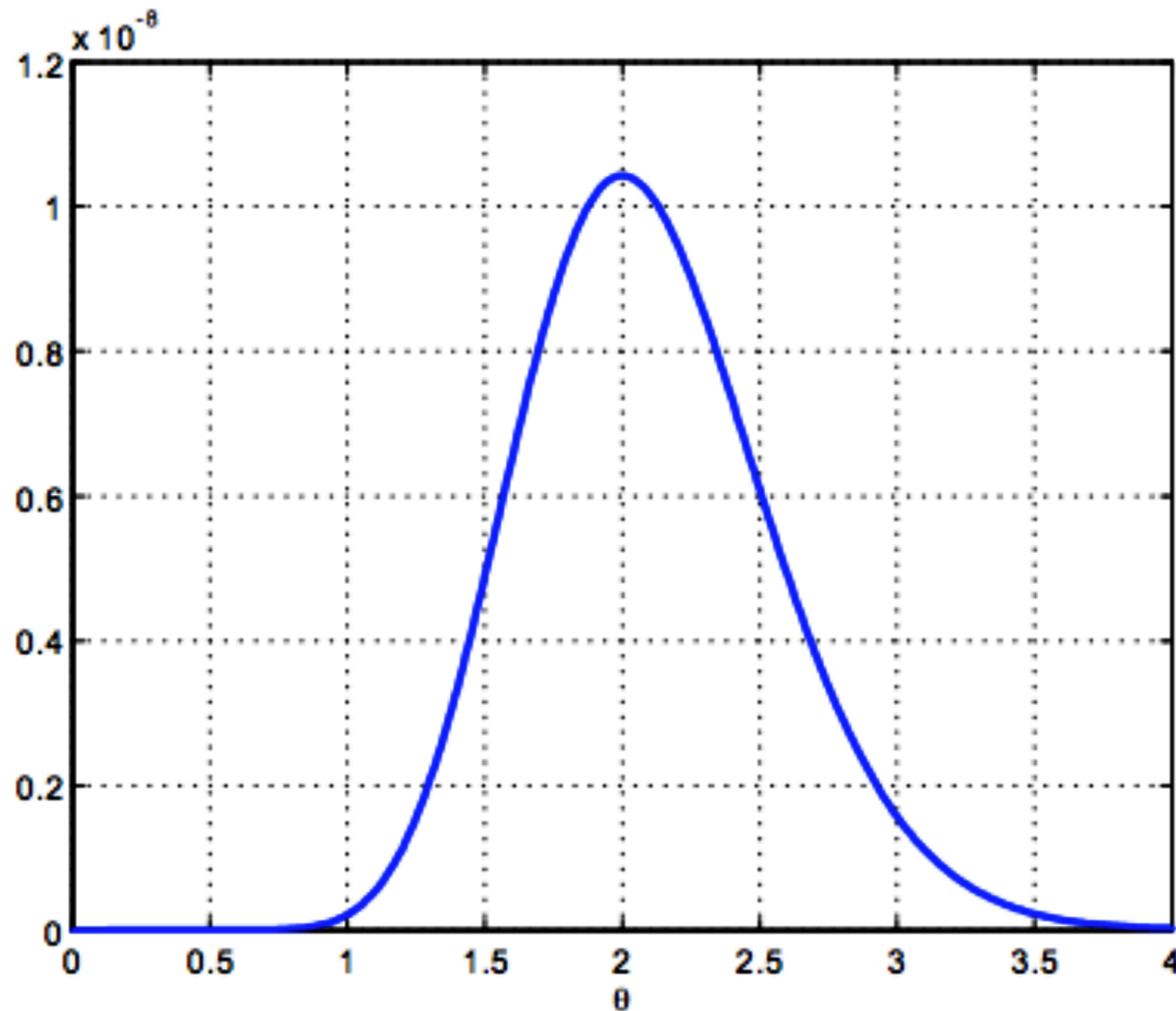
$$L_m(\theta; x^1, \dots, x^m) = \Pr((X^1 = x^1) \cap \dots \cap (X^m = x^m))$$

$$L_m(\theta; x^1, \dots, x^m) = \frac{e^{-10\theta} \theta^{20}}{207360}$$

Question: What value of θ would make this sample most probable?

The principle of maximum likelihood

This Figure plots the function $L_m(\theta; x^1, \dots, x^m)$ for various values of θ . It has a single mode at $\theta = 2$, which would be the Maximum Likelihood Estimate (MLE) of θ



The principle of maximum likelihood

Consider maximizing the likelihood function $L_m(\theta; x^1, \dots, x^m)$ with respect to θ . Since the log function is monotonically increasing, we usually maximize

$$\ln L_m(\theta; x^1, \dots, x^m)$$

instead. In this case:

$$\hat{\theta} = \arg \max_{\theta} \ln L_m(\theta; x^1, \dots, x^m)$$

The principle of maximum likelihood

$$\ln L_m(\theta; x^1, \dots, x^m) = -\theta m + \ln(\theta) \sum_{i=1}^m x^i - \ln \left(\prod_{i=1}^m x^i! \right)$$

$$\frac{dL_m(\theta; x^1, \dots, x^m)}{d\theta} = -m + \frac{1}{\theta} \sum_{i=1}^m x^i$$

$$\frac{d^2 L_m(\theta; x^1, \dots, x^m)}{d\theta^2} = -\frac{1}{\theta^2} \sum_{i=1}^m x^i \quad \Rightarrow \text{concave!!}$$

Maximizing a concave function is equivalent to minimizing a convex function!!

$$\Rightarrow \hat{\theta} = \arg \min_{\theta} -\ln L_m(\theta; x^1, \dots, x^m)$$

The principle of maximum likelihood

Therefore,

$$-\frac{\partial L_m(\theta; x^1, \dots, x^m)}{\partial \theta} = m - \frac{1}{\theta} \sum_{i=1}^m x^i = 0$$

$$\Rightarrow \hat{\theta} = \frac{1}{m} \sum_{i=1}^m x^i$$

This is called the **maximum likelihood estimate (estimator)**

The principle of maximum likelihood

- ▶ The maximum likelihood **estimator (random variable)** is

$$\hat{\theta} = \frac{1}{m} \sum_{i=1}^m X^i$$

- ▶ The maximum likelihood **estimate (realization)** is

$$\hat{\theta} \equiv \hat{\theta}(x) = \frac{1}{m} \sum_{i=1}^m x^i.$$

Given the sample $\{5, 0, 1, 1, 0, 3, 2, 3, 4, 1\}$, we have $\hat{\theta}(x) = 2$.

The principle of maximum likelihood

Continuous variables

- ▶ The reference to the probability of observing the given sample is exact in a continuous distribution, since a particular sample has probability zero. Nonetheless, the principle is the same.
- ▶ The likelihood function then corresponds to the pdf associated to the **joint distribution** of $(X^1, X^2, \dots, X^m)$ evaluated at the point $(x^1, x^2, \dots, x^m)$

$$L_m(\theta; x^1, \dots, x^m) = f_{X^1, \dots, X^m}(x^1, \dots, x^m; \theta)$$

The principle of maximum likelihood

Continuous variables

- If the random variables $\{X^1, \dots, X^m\}$ are i.i.d., then we have

$$L_m(\theta; x^1, \dots, x^m) = \prod_{i=1}^m f_X(x^i; \theta)$$

where $f_X(x^i; \theta)$ denotes the pdf of the marginal distribution of X (or X^i since all the variables have the same distribution).

- The values of the parameters that maximize $L_m(\theta; x^1, \dots, x^m)$ or its log are the maximum likelihood estimates, denoted $\hat{\theta}(x)$.

The likelihood

Definition (Likelihood function)

The likelihood function is defined to be $L_m : \Theta \times \mathbb{R}^m \rightarrow \mathbb{R}_+$

$$L_m(\theta; x^1, \dots, x^m) = \prod_{i=1}^m f_X(x^i; \theta)$$

Definition (log Likelihood function)

The log likelihood function is defined to be $l_m : \Theta \times \mathbb{R}^m \rightarrow \mathbb{R}_+$

$$l_m(\theta; x^1, \dots, x^m) = \sum_{i=1}^m \ln f_X(x^i; \theta)$$

Logarithmically concave function

Definition

$f: \mathbb{R}^n \rightarrow \mathbb{R}_+$ is logarithmically concave (or log-concave for short) if its domain is a convex set, and if it satisfies the inequality

$$f(\alpha x + (1 - \alpha)y) \geq f(x)^\alpha f(y)^{1-\alpha}$$

for all x, y in the domain and $0 < \alpha < 1$.

If f is strictly positive, this is equivalent to saying that the logarithm of the function, $\log f$ is concave

$$\log f(\alpha x + (1 - \alpha)y) \geq \alpha \log f(x) + (1 - \alpha) \log f(y).$$

Logarithmically concave function

As it happens, many common probability distributions are log-concave!!

- ▶ Normal distribution
- ▶ Uniform distribution
- ▶ Poisson distribution
- ▶ Exponential distribution
- ▶ Logistic distribution
- ▶ χ^2 distribution
- ▶

Univariate Gaussian Distribution

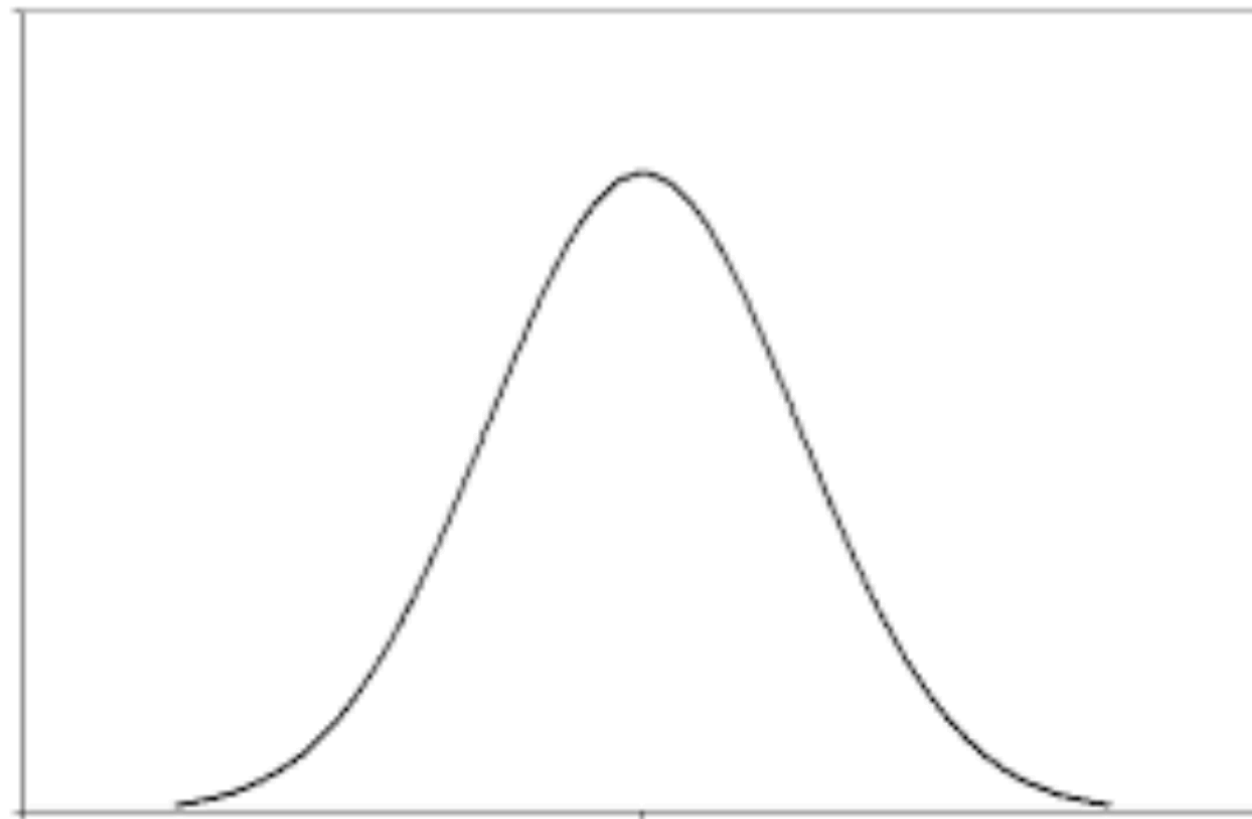
The probability density function (pdf) of a Gaussian distribution is given by

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \quad x \sim \mathcal{N}(\mu, \sigma^2)$$

where μ is the mean or center of mass and σ^2 is the variance

μ : unknown, needs to be estimated

σ^2 : known, but we don't care about it



The likelihood of Gaussian distribution

We consider a sample $\{X^1, \dots, X^m\}$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$ and denote the realization $\{x^1, \dots, x^m\}$ or simply x . Then we have

$$L_m(\mu, \sigma; x) = \prod_{i=1}^m \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{(x^i - \mu)^2}{2\sigma^2} \right)$$
$$l_m(\mu, \sigma; x) = \ln(\sigma^2 2\pi)^{-m/2} + \left(-\frac{1}{2\sigma^2} \sum_{i=1}^m (x^i - \mu)^2 \right)$$

so we have

$$l_m(\mu, \sigma; x) = -\frac{m}{2} \ln \sigma^2 - \frac{m}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^m (x^i - \mu)^2$$

MLE v.s. LS

Consider the linear regression model:

$$y = \mathbf{x}^T \theta + \varepsilon$$

where $\theta = [\theta_1, \dots, \theta_n]^T$.

We assume that ε_i are i.i.d. with $\varepsilon^i \sim \mathcal{N}(0, \sigma^2)$. Then the conditional distribution of Y_i given that $X^i = \mathbf{x}^i$ is

$$Pr(Y_i | \mathbf{x}^i) \sim \mathcal{N}(\mathbf{x}^{iT} \theta, \sigma^2)$$

and

$$\begin{aligned} L_m(\mathbf{x}^i \theta; y|x) &= \prod_{i=1}^m f_{Y|x}(y^i | \mathbf{x}^i; \mathbf{x}^{iT} \theta) \\ &= \prod_{i=1}^m \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{(y^i - \mathbf{x}^{iT} \theta)^2}{2\sigma^2} \right) \end{aligned}$$

MLE v.s. LS

The log MLE is given by

$$\begin{aligned} l_m(\mathbf{x}^i \theta; y|x) &= \sum_{i=1}^m \ln f_{Y|x}(y^i | \mathbf{x}^i; \mathbf{x}^{iT} \theta) \\ &= \sum_{i=1}^m \ln \frac{1}{\sigma \sqrt{2\pi}} \exp \left(-\frac{(y^i - \mathbf{x}^{iT} \theta)^2}{2\sigma^2} \right) \\ &= -\frac{m}{2} \ln \sigma^2 - \frac{m}{2} \ln(2\pi) - \frac{1}{\sigma^2} \frac{1}{2} \|\mathbf{X}\theta - \mathbf{y}\|^2 \end{aligned}$$

MLE v.s. LS

The log MLE is given by

$$l_m(\mathbf{x}^i \theta; y|x) = -\frac{m}{2} \ln \sigma^2 - \frac{m}{2} \ln(2\pi) - \frac{1}{\sigma^2} \frac{1}{2} \|\mathbf{X}\theta - \mathbf{y}\|^2$$

To find the MLE of θ , we set the gradient to zero

$$\frac{\partial}{\partial \theta} l_m(\mathbf{x}^i \theta; y|x) = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{X} \theta - \mathbf{X}^T \mathbf{y}) = 0$$

$$\Rightarrow \hat{\theta} = (\mathbf{X}^T \mathbf{X})^+ \mathbf{X}^T \mathbf{y}$$

Conclusion: Under the **previous probabilistic assumptions** on the data, least-squares regression corresponds to finding the maximum likelihood estimate of θ !!

Mean Absolute Error (MAE)

- ▶ **Laplace noise** In case of linear regression, suppose data set $(\mathbf{X}, \mathbf{y})$ is generated according to

$$y = x^T \boldsymbol{\theta} + \epsilon$$

where $\mathbf{X} \in \mathbb{R}^{m \times n}$, $\mathbf{X} = [x_1, x_2, \dots, x_n]^T$, $\mathbf{y} \in \mathbb{R}^m$ and $\boldsymbol{\theta} \in \mathbb{R}^n$.

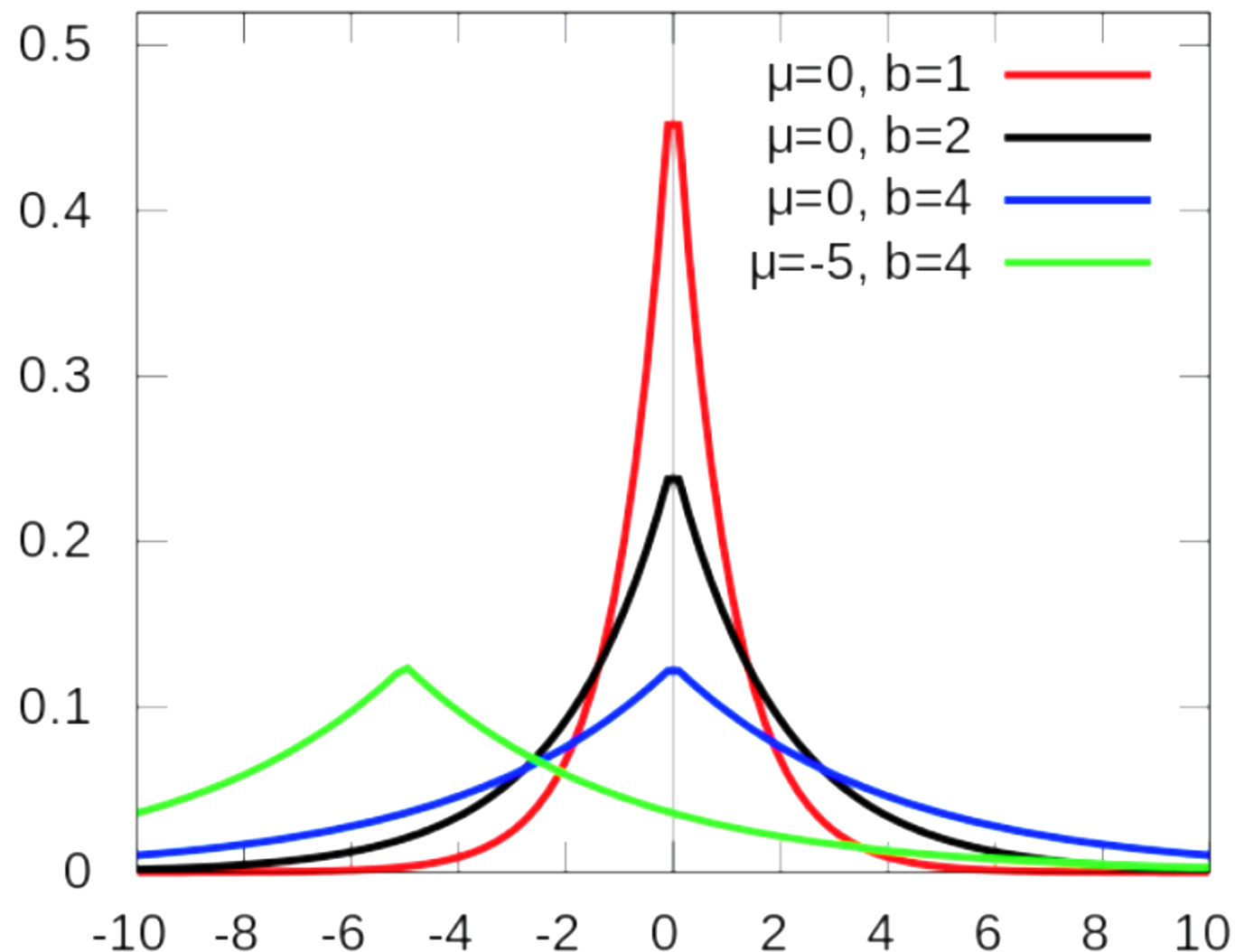
- ▶ Assume ϵ is the Laplace noise with variance $2b^2$, i.e., $\epsilon \sim \text{Laplace}(0, b)$.

Mean Absolute Error (MAE)

For a data point $(\mathbf{x}_i, y_i)$, it follows that $y_i \sim \text{Laplace}(\mathbf{x}_i^T \boldsymbol{\theta}, b)$ with conditional density function

$$p(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}) = \frac{1}{2b} \exp \left\{ -\frac{|\mathbf{x}_i^T \boldsymbol{\theta} - y_i|}{b} \right\}.$$

Probability density function



Mean Absolute Error (MAE)

- Suppose the data points are generated i.i.d., so that the given $\mathbf{X}$ (with parameter $\boldsymbol{\theta}$) the likelihood of observing $\mathbf{y}$ is given by

$$L(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \prod_{i=1}^m \frac{1}{2b} \exp \left\{ -\frac{|\mathbf{x}_i^T \boldsymbol{\theta} - y_i|}{b} \right\}.$$

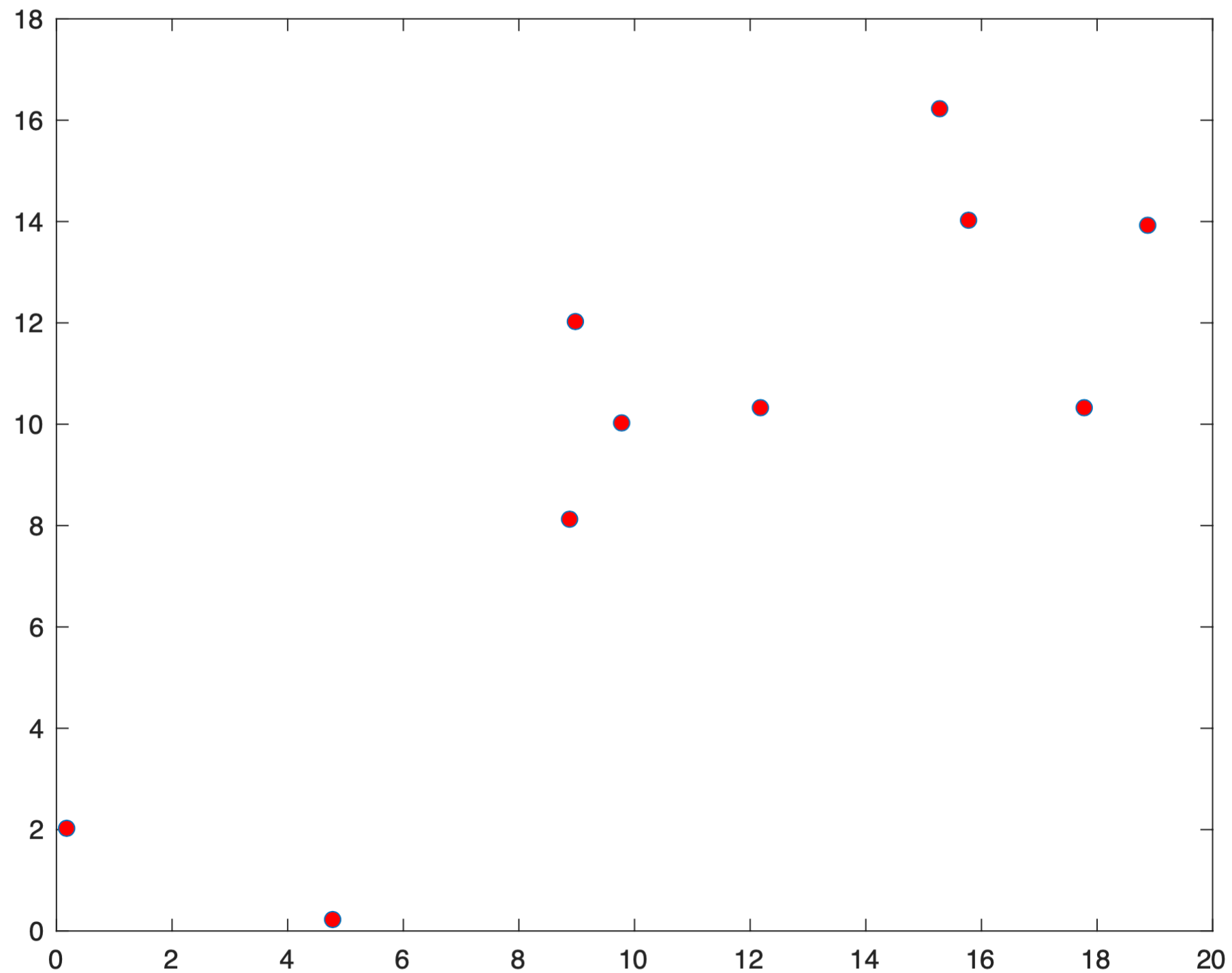
- The “best” parameter $\boldsymbol{\theta}$ should be chosen as the one that maximize this likelihood, which can be given by

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \arg \max_{\boldsymbol{\theta}} L(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \\ &= \arg \max_{\boldsymbol{\theta}} \ln L(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \\ &= \arg \min_{\boldsymbol{\theta}} -\ln L(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \\ &= \arg \min_{\boldsymbol{\theta}} m \ln 2b + \frac{1}{2\delta^2} \sum_{i=1}^m |\mathbf{x}_i^T \boldsymbol{\theta} - y_i| \\ &= \arg \min_{\boldsymbol{\theta}} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_1 \end{aligned}$$

4. Effects of MSE

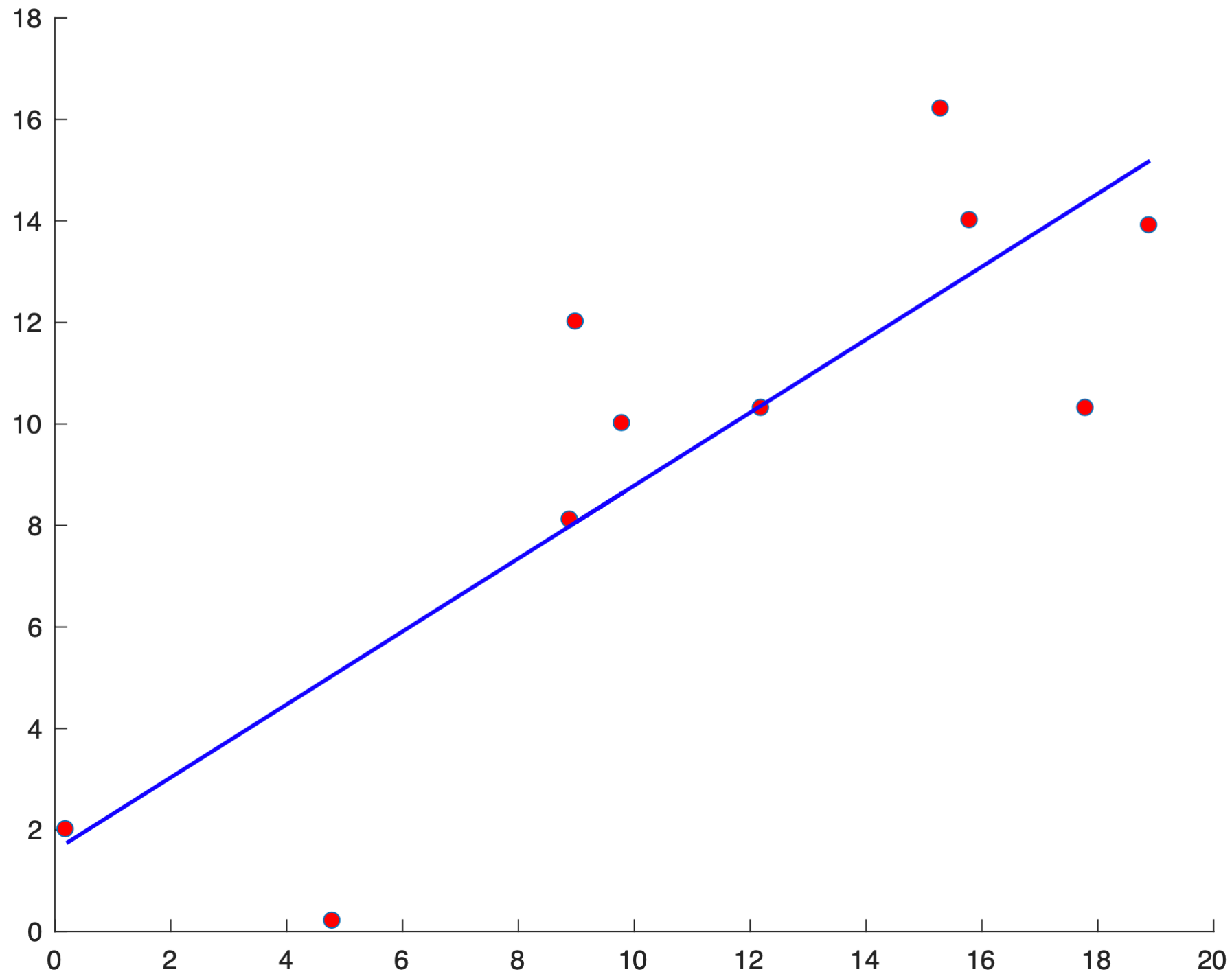
Effects of MSE

Sensitivity to outliers



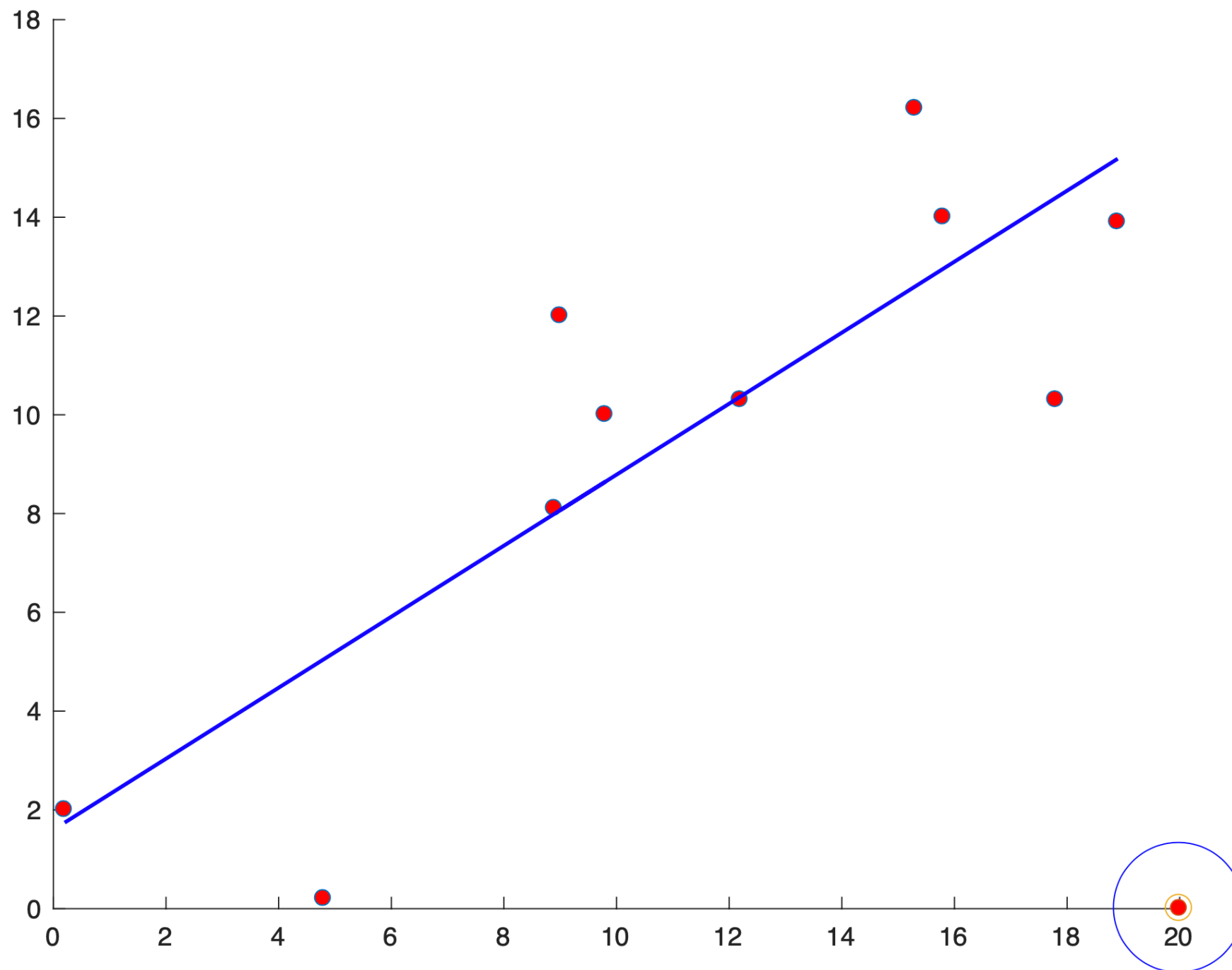
Effects of MSE

Sensitivity to outliers



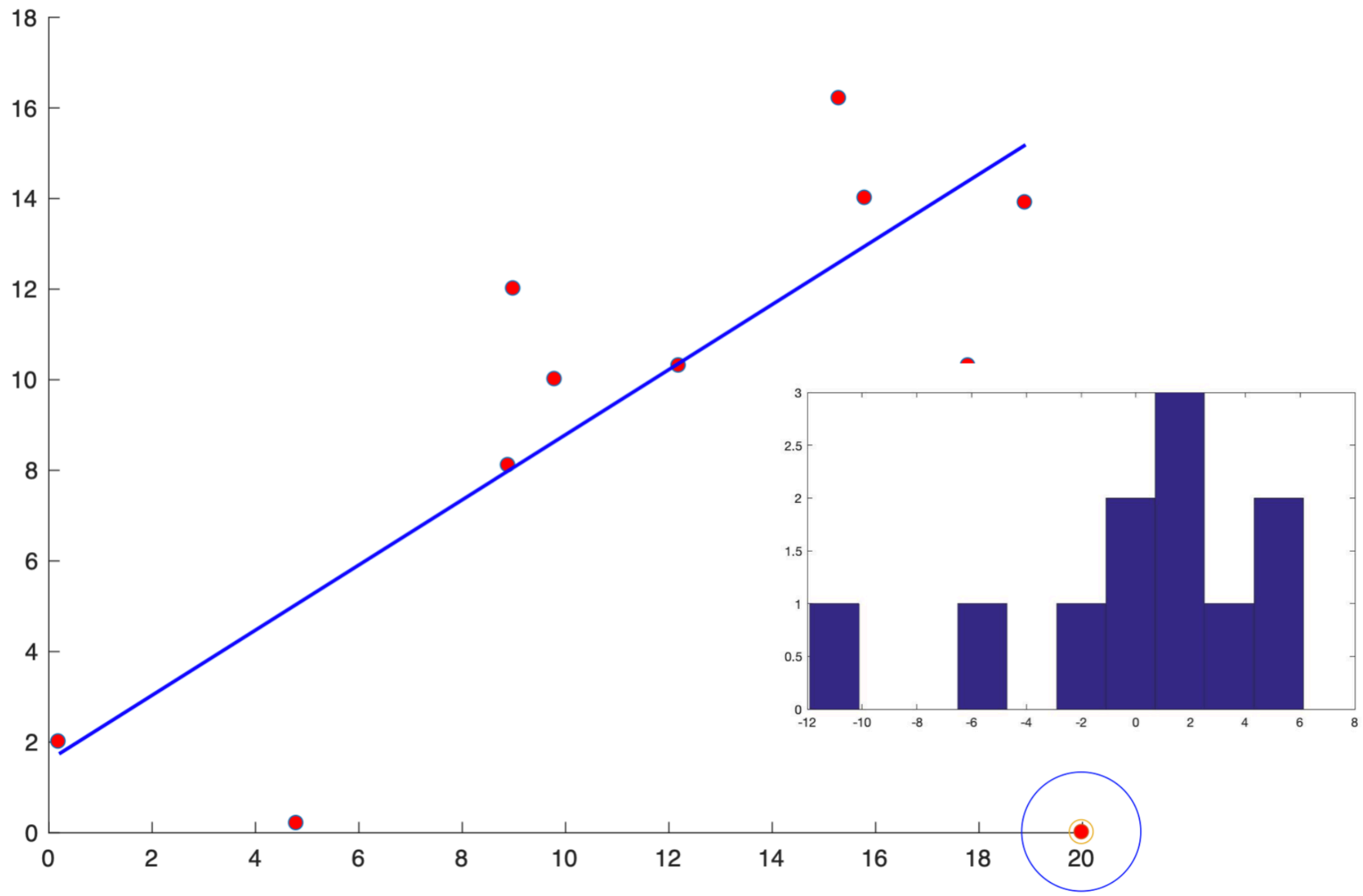
Effects of MSE

Sensitivity to outliers



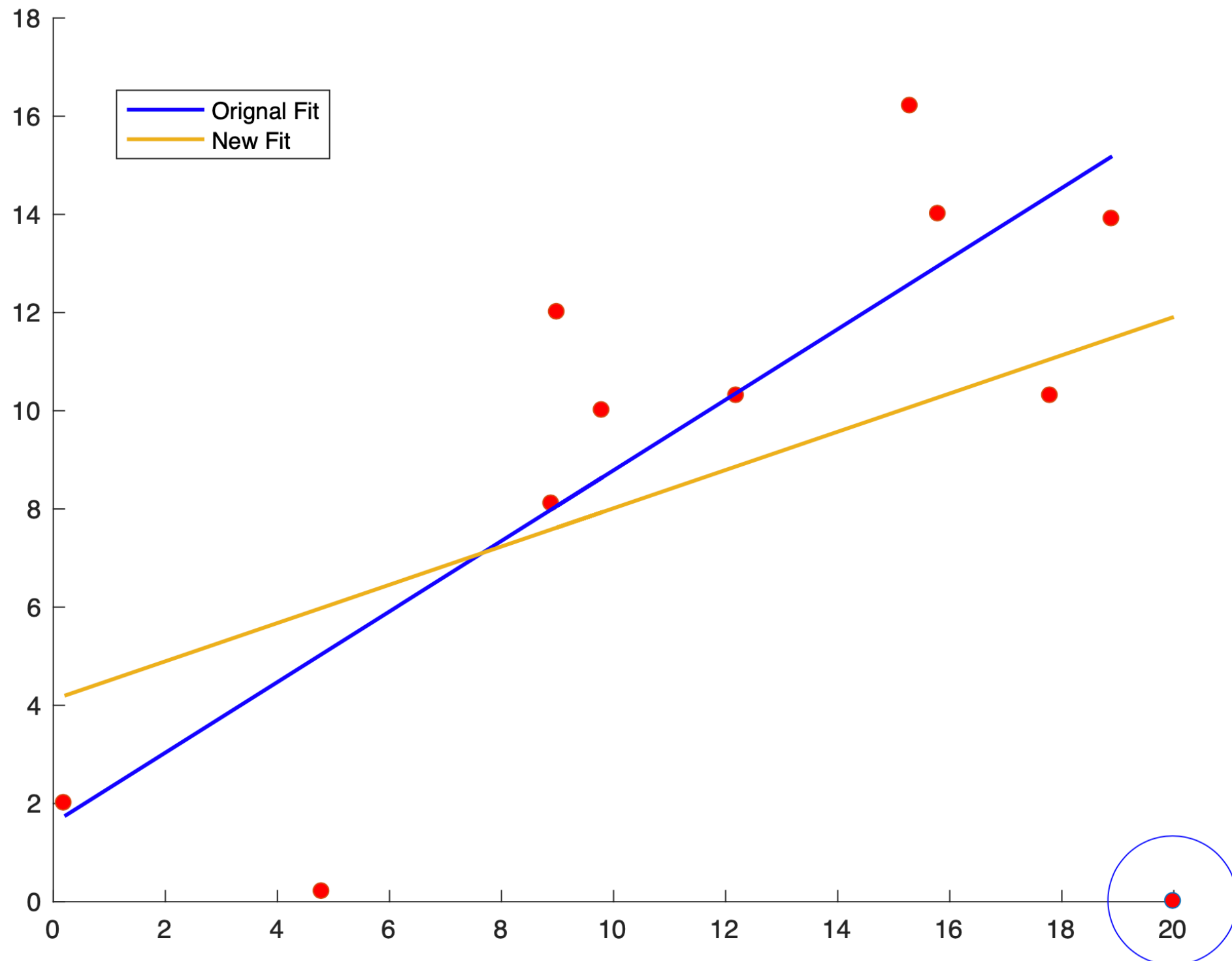
16^2 cost for this one datum. Heavy penalty for large errors

Effects of MSE



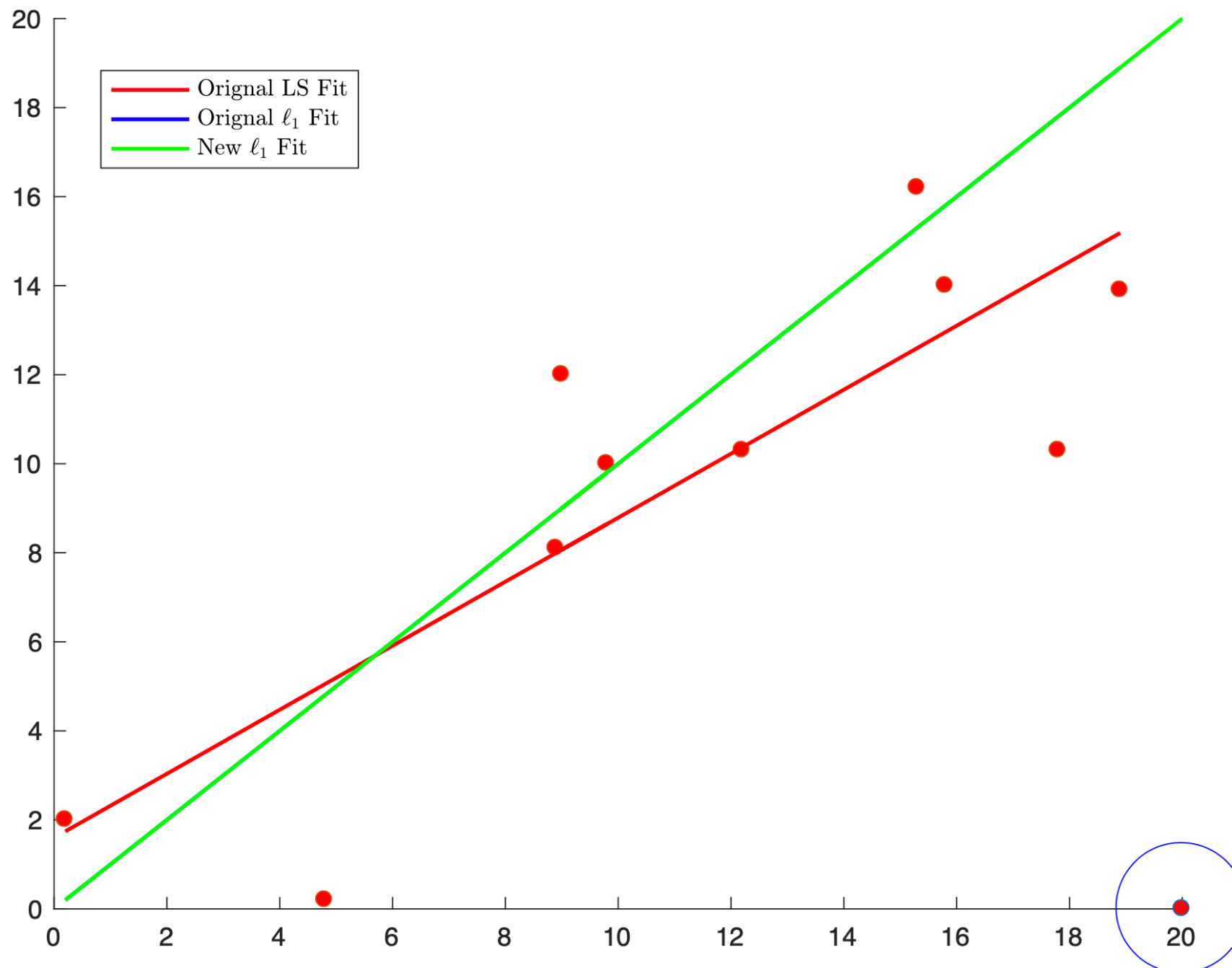
Effects of MSE

Sensitivity to outliers

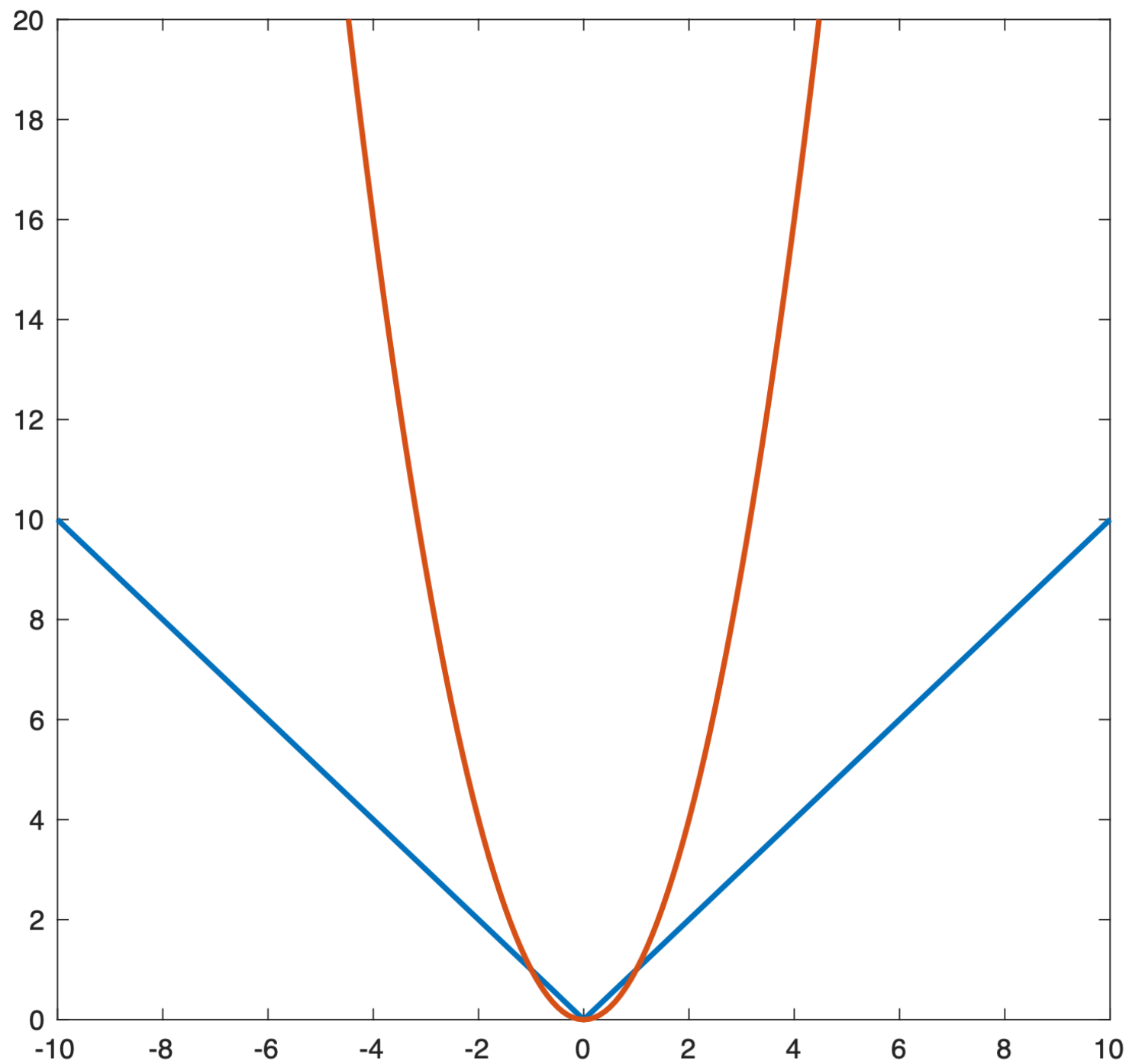


ℓ_1 Error Regression (MAE)

$$J(\theta) = \sum_{i=1}^m |y^i - \mathbf{x}^i \theta|$$



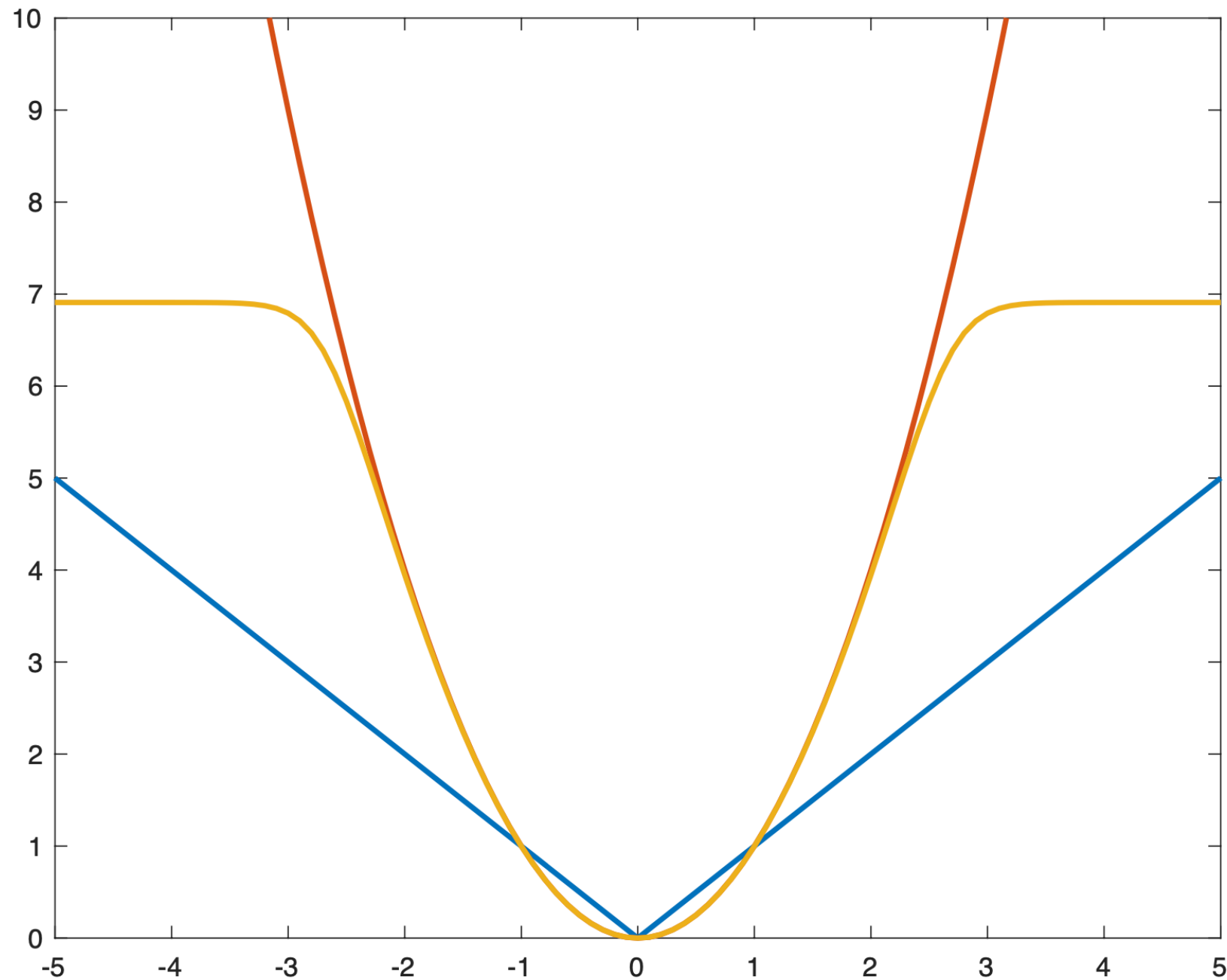
Cost/loss functions



$$\ell_2: (y - \hat{y})^2 \text{ (MSE)}$$

$$\ell_1: |y - \hat{y}| \text{ (MAE)}$$

Cost/loss functions



$$\ell_2: (y - \hat{y})^2 \text{ (MSE)}$$
$$\ell_1: |y - \hat{y}| \text{ (MAE)}$$

Something else...

log error: $c - \log(\exp(-(y - \hat{y})^2) + c)$